FWMS — Phase I: Business Process Modeling
Author: Nkurunziza Aimable
Student ID:29692
Business: Kwizera Decor Interior Design

TASK 1: BUSINESS PROCESS DEFINITION

Business Process Selected: Custom Furniture Order Fulfillment at Kwizera Decor Interior Design.

Purpose: This process covers how a customer's furniture need is handled — from first contact to final delivery and installation. It begins when a customer comes to our showroom and either finds a product they are interested in or shows us a photo of the product they want. We then negotiate the price, and if the customer accepts, our boss prepares a quotation, goes to the market to buy the required materials, and production begins. The process ends when we deliver the furniture to the customer's location and the customer pays the remaining 50% balance, since 50% was already paid at the start.

Objectives: The process aims to:

(a) clearly understand the customer's needs so that we build exactly what they want.

(b) produce an accurate quotation so that materials are not miscalculated and the business does not face unexpected losses due to disorganization.

(c) deliver the finished product safely and on time, ensuring it reaches the customer's location without damage.

Expected Outcome: A satisfied customer receiving furniture that matches the approved design, with full payment received and materials properly accounted for.

Problem it addresses: Currently at Kwizera Decor, all order progress is managed by the boss from memory, with no written records. This causes materials to be bought twice, orders to be delayed when there are many customers at once, customer complaints about agreed prices due to missing records, and no proper tracking of showroom products, broken items, materials, or machines. This process exists to bring structure, accountability, and visibility to these activities.

Why it is important: This is the core revenue-generating process of the business — every custom order runs through it. A well-modeled process is the foundation for the database that will later replace the "memory-based" system with reliable records.

TASK 2: ACTORS AND RESPONSIBILITIES

Numbers	Actor	Type	Responsibility in the Process

1)	Customer	External	Requests furniture (walks in showroom, Makes a phone, WhatsApp) and shows photos or selects from showroom,negotiates price, approves design and quotation; pays 50% deposit and 50% balance after
2)	Boss / Designer (Kwizera Enock)	Internal – Management	Measures the space; creates designs and sketches, prepares quotations, buys materials from the market, assigns work to carpenters, tracks overall progress (currently from memory), and after all receives all payments
3)	Showroom Attendant	Internal – Employee	Welcomes and receives all customers (reception + sales); shows products in the showroom; passes custom order requests to the boss (does NOT prepare quotations)
4)	Marketing Officer (Nkurunziza Aimable)	Internal – Employee	Advertises the business, attracts new customers, handles marketing communication
5)	Carpenters (Mechack, Elia, Danny, Keve + others)	Internal – Employees	Manufacture furniture according to the approved design, and reports progress to the boss
6)	Cleaner	Internal – Employee	Maintains cleanliness of the workshop and showroom

7)	Transport / Delivery Drivers	External – Ad hoc	Hired temporarily to deliver finished furniture to the customer's location
8)	Material Suppliers / Market Vendors	External	Provide raw materials (wood, fabric, hardware) purchased by the boss
9)	Installation Team	Internal – Employees	Install the delivered furniture at the customer's site
10)	FWMS Database System *(future)*	System	Will store orders, quotations, materials, payments, and worker assignments; replaces the boss's memory-based tracking; supports reporting and decision-making

TASK 3: BUSINESS PROCESS DIAGRAM

The business process model for Kwizera Decor was created using draw.io (diagrams.net). The diagram uses a BPMN Swimlane format to clearly show the interactions between the Customer, Showroom Attendant, Boss/Designer, Carpenters, and the Delivery team. It illustrates the start and end events, decision gateways, and the flow of information.
Link to the published interactive diagram:

Click here to view the interactive BPMN Diagram

TASK 4: DATA-GENERATING ACTIVITIES (CRUD)

To support the future database design, the diagram identifies the activities that generate or interact with data, marked with CRUD tags:

1) **Create [C]:** Customer requests furniture, Measures space, Creates design & sketch, Prepares quotation, Buys materials, Hires transport. These activities generate new records in the system.

2) **Retrieve/Read [R]:** Shows products, In showroom?. These activities consult existing data (such as showroom inventory) to make decisions.

3) **Update [U]:** Assigns work, Manufactures furniture, Reports progress, Installs furniture, Receives furniture and pays balance. These activities modify existing records (e.g., changing an order status from "in production" to "completed").

TASK 5: ONE-PAGE PROCESS EXPLANATION

The custom furniture order fulfillment process at Kwizera Decor begins when a customer contacts the business through a walk-in visit to the showroom, a phone call, or WhatsApp. The process is initiated by the Showroom Attendant, who welcomes the customer and shows them the available products. A decision point occurs here: if the customer finds a product they like in the showroom, the attendant retrieves the inventory information to confirm availability, and the customer proceeds directly to payment and delivery arrangements. However, if the requested item is not available in the showroom, the process moves into the custom design phase, and the customer is directed to the Boss/Designer.

At this stage, the Boss discusses the customer's needs and reviews any photos the customer provides. The Boss then measures the space, creates a design and sketch, and prepares a quotation. This quotation is sent back to the customer for review. A key decision point occurs here: if the customer does not accept the price, the process loops back to the discussion phase so that the needs and design can be revised. If the customer accepts the quotation, they pay a 50% deposit, and the process moves into production. The Boss then purchases the required materials from the market and assigns the work to the carpenters.

Once the carpenters begin manufacturing the furniture, they update the production status and report progress to the Boss. After the furniture is completed, the Boss arranges for transport and delivery to the customer's location, and the installation team installs the furniture. Finally, the customer receives the furniture and pays the remaining 50% balance, and the order is marked as complete.

This business process is important because it ensures that every custom order is tracked from the initial request to final delivery. It shows how information flows between the customer, the showroom, the designer, the carpenters, and the delivery team. It also demonstrates how

decisions are made (such as whether to buy from the showroom or build a custom piece, and whether the customer accepts the quotation). By modeling this process, the foundation is laid for a database system that will replace the current memory-based tracking, solving problems such as materials being bought twice, order delays, and missing customer records.

REFERENCES

1. Kwizera Decor. (2026). Interior Design & Home Decoration in Rwanda. Retrieved from https://www.kwizerdecor.com/
2. Object Management Group (OMG). (2011). Business Process Model and Notation (BPMN) Version 2.0. Retrieved from https://www.omg.org/spec/BPMN/2.0/
3. Draw.io (Diagrams.net). (2026). Security and Privacy. Retrieved from https://www.drawio.com/doc/faq/security-privacy